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Assignment Cover Sheet

All assignments are to be submitted, on or before the due date of the assignment.

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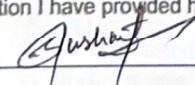
- ☐ Master of Business Administration
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Semester: (Please darken ☒ one)① ② ③ ☒ ⑤ ⑤ ⑥Subject Code: WLC-234 Subject Title: Wireless CommunicationAssignment Title: IIAssignment Due Date: 14/09/2026 Instructor's Name: Sauju ShresthaStudent Name(s): SUSHANT PYAKUREL

Student Number or Group Name: _____ Word Count: _____

Your assignment should meet ALL the following requirements:(Please confirm by ticking ☒ each item before submitting your assignment)

- ☒ Assignment is in an easy to read format (12pt font, e.g. Arial, Calibri or Times Roman, with 1.5 spacing)
☒ Assignment is referenced properly
☒ Pages are numbered
☒ Assignment is stapled at the top left hand corner
☒ Declaration below is completed and signed
☒ I have kept a copy of this assignment

DO NOT submit your assignment in any kind of cover, except with the prior approval of the Instructor.**ALL FORMS OF PLAGIARISM AND UNAUTHORISED COLLUSION** are regarded as serious academic misconduct by KFA Business School and may result in penalties, including failure of the unit and possible disciplinary action. If you are in doubt, consult your Instructor.**DECLARATION (Must be signed):** Except where appropriately acknowledged, this assignment is my own work, has been expressed in my own words and has not previously been submitted for assessment. The information I have provided herein is true and correct.Signed: Date: 14th Sept 2026

WIRELESS COMMUNICATION (WLC-234)

Assignment 2

Submission Deadline: 14th September 2026

- 1) If 0 dBm is equal to 1 mW (10^{-3} W) over a 50 Ω load; express 10 W in units of dBm.
- 2) A total of 24 MHz of bandwidth is allocated to a particular FDD cellular telephone system that uses two 30kHz simplex channels to provide full duplex voice and control channels. Assume each cell phone user generates 0.1 Erlangs of traffic. Assume Erlang B is used.
 - a) Find the number of channels in each cell for a four-cell reuse system.
 - b) If each cell is to offer capacity that is 90% of perfect scheduling, find the maximum number of users that can be supported per cell where omnidirectional antennas are used at each base station.
 - c) What is the blocking probability of the system in (b) when the maximum number of users are available in the user pool?
 - d) If each new cell now uses 120° sectoring instead of omnidirectional for each base station, what is the new total number of users that can be supported per cell for the same blocking probability as in (c)?
 - e) If each cell covers five square kilometers, then how many subscribers could be supported in an urban market that is 50 km x 50 km for the case of omnidirectional base station antennas?
 - f) If each cell covers five square kilometers, then how many subscribers could be supported in an urban market that is 50 km x 50 km for the case of 120° sectored antennas?
- 3) If 20 MHz of total spectrum is allocated for a duplex wireless cellular system and each simplex channel has 25 kHz RF bandwidth, find: (a) the number of duplex channels. (b) the total number of channels per cell site, if $N = 4$ cell reuse is used.
- 4) A certain area is covered by a cellular radio system with 84 cells and a cluster size N . 300 voice channels are available for the system. Users are uniformly distributed over the area covered by the cellular system, and the offered traffic per user is 0.04 Erlang. Assume that blocked calls are cleared and the designated blocking probability is $P_b = 1\%$.
 - a) Determine the maximum carried traffic per cell if cluster size $N = 4$ is used. Repeat for cluster sizes $N = 7$ and 12.
 - b) Determine the maximum number of users that can be served by the system for a blocking probability of 1% and cluster size $N = 4$. Repeat for cluster sizes $N = 7$ and 12.
- 5) The GSM TDMA system uses a 270.833 kbps data rate to support eight users per frame.
 - a) What is the raw data rate provided for each user?
 - b) If guard time, ramp-up time, and synchronization bits occupy 10.1 kbps, determine the traffic efficiency for each user.

- 6) Consider a cellular network with 12 cells arranged in a hexagonal pattern. Each cell needs to be assigned a frequency channel to support communication for mobile devices. The available frequency channels are numbered from 1 to 20.
 - a) Determine the total number of channels available.
 - b) Assign the channel using channel assignment strategies.
- 7) Features collectively make DECT a reliable and versatile wireless communication solution, well-suited for both residential and business environments. Emphasize on features, frequency of DECT.
- 8) Assume a system of 32 cells with a cell radius of 1.6 km, a total of 32 cells, a total frequency bandwidth that supports 336 traffic channels and a reuse factor of $N = 7$. If there are 32 total cells, what geographic area is covered, how many channels are there per cell, and what is the total number of concurrent calls that can be handled?
- 9) ISDN has been more widely accepted as a network solution, there are various types of ISDN; clarify it, include speed, uses.
- 10) Consider GSM which is a TDMA/FDD system that uses 20MHz for the forward link, which is broken into radio channels of 20KHz. If 6 speech channels are supported on a single radio channel, and if no guard band is assumed. Find the number of simultaneous users that can accommodate in GSM.
- 11) Describe the key features and functions of Signaling System No. 7 (SS7) in telecommunications networks. Explain how SS7 facilitates call setup, routing, and termination processes. Discuss the advantages of using SS7 in comparison to in-band signaling techniques.
- 12) On average, there are 1800 new calls in an hour, and the average holding time is 3 minutes. Calculate the traffic intensity.
- 13) If a particular FDD cellular telephone system has a total bandwidth of 33 MHz, and if the phone system uses two 25 KHz simplex channels to provide full duplex voice and control channels. Compute the number of channels per cell if $N = 4, 7, 12$.
- 14) The US Digital Cellular TDMA system uses a 48.6 kbps data rate to support three users per frame. Each user occupies two of the six time slots per frame. What is the raw data rate provided for each user?

① Soln:

$$\text{Power (dBm)} = 10 \log_{10} (P/1 \text{ mW})$$

$$\begin{aligned}\text{Power (dBm)} &= 10 \log_{10} (10 \text{ W} / 10^{-3} \text{ W}) \\ &= 10 \log_{10} (10^4) \\ &= 40 \text{ dBm}\end{aligned}$$

② Soln:

$$\begin{aligned}\text{Total Duplex channels } C &= 24 \text{ MHz} / (2 \times 30 \text{ kHz}) \\ &= 24000 / 60 \\ &= 400 \text{ channels}\end{aligned}$$

③ Channels per cell
($N=4$):

$$C_{\text{cell}} = 400/4 = 100 \text{ channels/cell}$$

④ Maximum offered traffic at 90% capacity:

$$\begin{aligned}A_{\text{max}} &= 0.90 \times 100 \\ &= 90 \text{ Erlangs}\end{aligned}$$

$$\text{Users/cell } U = 90/0.1 = 900 \text{ users/cell}$$

⑤ Blocking probability using Erlang B ($C=100, A=90$):

$$\begin{aligned}P_b &= [90^{100} / 100!] / \sum_{k=0}^{100} [90^k / k!] \\ &\approx 0.0232 (2.32\%)\end{aligned}$$

④ For 120° sectoring (3 sectors/cell):

$$C_{\text{sector}} = 100/3 \approx 33 \text{ channels}$$

For $C = 33$ and $P_b = 2.32\%$, $A_{\text{sector}} \approx 26.35 \text{ Erlangs}$

$$U_{\text{sector}} = 26.35/0.1 \approx 263 \text{ users}$$

$$U_{\text{cell}} = 3 \times 263 = 789 \text{ users/cell}$$

$$\textcircled{e} \text{ Market area} = 50 \times 50 = 2500 \text{ km}^2; \text{ cell} = \frac{2500}{5} \\ = 500 \text{ cells}$$

$$\text{Total subscribers} = 500 \times 900 = 450,000 \text{ subscribers}$$

⑤ For 120° sectored antennas:

$$\text{Total subscribers} = 500 \times 789 \\ = 394,500 \text{ subscribers}$$

⑥ Solⁿ:

$$\text{Duplex channel bandwidth} = 2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

⑦ (a) Number of duplex channels:

$$C = 20,000 \text{ kHz} / 50 \text{ kHz} \\ = 400 \text{ channels}$$

⑧ (b) channels per cell ($N=4$):

$$K = 400/4 = 100 \text{ channels/cell}$$

④ Soln:

Total cells = 84, total channels = 300,

$$A_u = 0.04 \text{ Erlangs}, P_b = 0.01 (1\%).$$

(a) Carried traffic ($A_c = A(1 - P_b)$):

$$N=4: C = 300/4 = 75 \Rightarrow A \approx 62.11 \Rightarrow A_c = 62.11 \times 0.99 \\ = 61.49 \text{ Erlangs/cell}$$

$$N=7: C = \lceil 300/7 \rceil = 42 \Rightarrow A \approx 31.51 \Rightarrow A_c = 31.55 \times 0.99 \\ = 31.24 \text{ Erlangs/cell}$$

$$N=12: C = 300/12 = 25 \Rightarrow A \approx 16.12 \Rightarrow A_c = 16.12 \times 0.99 \\ = 15.96 \text{ Erlangs/cell}$$

(b) Total system users ($U_{\text{total}} = A/A_u \times 84$):

$$N=4: U_{\text{cell}} = 62.11/0.04 \approx 1552 \Rightarrow U_{\text{total}} = 1552 \times 84 \\ = 130,368 \text{ users}$$

$$N=7: U_{\text{cell}} = 31.51/0.04 \approx 787 \Rightarrow U_{\text{total}} = 787 \times 84 \\ = 66,108 \text{ users}$$

$$N=12: U_{\text{cell}} = 16.12/0.04 \approx 403 \Rightarrow U_{\text{total}} = 403 \times 84 \\ = 33,852 \text{ users}$$

⑤ Soln:

(a) Raw data rate per user:

$$R_{\text{raw}} = 270.833 \text{ kbps} / 8 \\ = 33.854 \text{ kbps}$$

(b) Traffic efficiency:

$$\text{Useful rate} = 33.854 - 10.1 = 23.754 \text{ kbps}$$

$$\text{Efficiency} = (23.754/33.854) \times 100\% \approx 70.16\%$$

⑥ Soln:

① Total channels available = 20 (numbered 1 through 20).

② Channel assignment strategies:

- Fixed Channel Assignment (FCA): Channels are divided into fixed sets per cluster. For $N=4$ each group gets 5 channels; e.g. Group 1 gets $\{1, 5, 9, 13, 17\}$. Co-channel cell reuse the exact same set.
- Dynamic Channel Assignment (DCA): Channels are not permanently assigned to specific cells. All 20 channels stay in a central pool and are assigned on demand based on traffic and interference conditions.

⑦ DECT (Digital Enhanced Cordless Telecommunications):

Frequency band: 1880-1900 MHz (Europe/Global) and 1920-1930 MHz (US DECT 6.0).

Key features:

- Dynamic channel selection (DCS): Automatically selects the clearest channel to avoid interference.
- High security: Uses DSAA authentication and DSEC voice encryption.
- Dedicated spectrum: Prevents interference from Wi-Fi networks.
- Seamless handoff: Allows users to roam between cells without losing connection.

⑧ Soln:

Given:

Total cells = 32, $R = 1.6$ km, total channels = 336, $N = 7$.

$$\text{cell area} = (3\sqrt{3}/2) R^2 \approx 2.598 \times (1.6)^2 \approx 6.65 \text{ km}^2$$

$$\text{Total area covered} = 32 \times 6.65 \text{ km}^2 \approx 212.8 \text{ km}^2$$

$$\text{channels per cell} = 336/7 = 48 \text{ channels/cell}$$

$$\text{Total concurrent calls} = 32 \times 48 = 1536 \text{ concurrent calls}$$

⑨ ISDN types:

BRI (Basic Rate Interface): Structure: 2B + 1D channels (2 x 64 Kbps + 16 Kbps). Speed: 144 Kbps. User: residential internet, small office lines, backup connections.

PRI (Primary Rate Interface): Structure: T1 (23B + 1D) or E1 (30B + 1D). Speed: 1.544 Mbps (T1) / 2.048 Mbps (E1). Users: large enterprise PBX trunk lines, digital call centers.

B-ISDN (Broadband ISDN): Structure: ATM-based packet switching. Speed: 155 Mbps to 622 Mbps+. Users: high-speed telecommunication network backbones.

10. Soln:

$$\text{Number of radio channels} = \frac{20 \text{ MHz}}{20 \text{ kHz}}$$

$$= \frac{20,000}{20}$$

$$= 1000 \text{ radio channels.}$$

$$\text{Simultaneous users} = 1000 \times 6$$

$$= 6000 \text{ simultaneous users.}$$

11. SS7 (Signaling System No. 7):

Functions: Out-of-band signaling system that handles call setup, routing, teardown, SMS, and roaming lookup (HLR/VLR).

Call Setup Steps:

- IAM (Initial Address Message): Requests circuit reservation along the call path.
- ACM (Address Complete Message): Confirms the line is ringing at the destination.
- ANM (Answer Message): Connects the voice trunk when the called party picks up.
- REL/RLC (Release / Release Complete): Releases the circuit when the call ends.

Advantages over in-band signaling: Faster setup times; voice circuits are not blocked during setup; prevents fraud such as tone-manipulation attacks.

92) Soln:

Call rate $\lambda = 1800$ calls/hr,

holding time $H = 3 \text{ min} = 3/60 \text{ hr} = 0.05 \text{ hr}$.

Traffic intensity $A = \lambda \times H = 1800 \times 0.05$
 $= 90 \text{ Erlangs}$.

93) Soln:

Total Bandwidth $= 33 \text{ MHz} = 33,000 \text{ KHz}$;

duplex channel bandwidth $= 2 \times 25 \text{ KHz} = 50 \text{ KHz}$.

$C = 33000/50 = 660$ duplex channels

For $N=4$: $K = 660/4 = 165$ channels/cell.

For $N=7$: $K = \lfloor 660/7 \rfloor = 94$ channels/cell

For $N=12$: $K = 660/12 = 55$ channels/cell

94) Soln:

Raw data rate per user $= \frac{48.6 \text{ Kbps}}{3 \text{ users}}$

$= 16.2 \text{ Kbps}$